

An Explainable AI Framework for Intelligent Soil and Fertilizer Recommendation Using Random Forest, SHAP, and Fuzzy TOPSIS

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Abstract—Intelligent fertilizer recommendation is one of the crucial factors in precision agriculture for improving crop yield and ensuring soil sustainability. However, many machine learning-based systems are black-box models, and are not easily interpretable or trusted in real-world agricultural decision making. To overcome this, the present paper suggests an explainable AI framework for soil and fertilizer recommendation, which integrates the Random Forest classification, SHAP (SHapley Additive exPlanations) and Fuzzy TOPSIS. After class balancing using SMOTE, the system achieves a test accuracy of 87.30% and a macro F1 score of 74.96% on 10,000 soil-crop records for 19 agronomic features across seven fertilizer classes. SHAP TreeExplainer is used to explain the importance of different factors, including the levels of nitrogen, phosphorus, potassium and soil pH, both globally and locally. The importance scores from SHAP are converted to TFNs and then applied as weights in a Fuzzy TOPSIS multi-criteria decision making module to rank the different fertilizer alternatives according to their agronomic suitability. The framework also produces natural language explanations to make it more useful for farmers. The experimental results demonstrate that the proposed hybrid approach achieves comparable predictive accuracy and enhances the interpretability and transparency of decision-making in precision agriculture.

Index Terms—Fertilizer Recommendation, Explainable AI, Random Forest, SHAP (SHapley Additive exPlanations), Fuzzy TOPSIS, SMOTE (Synthetic Minority Over-sampling Technique), Precision Agriculture, Multi-criteria Decision Making.

I. INTRODUCTION

In many developing countries agriculture is still the mainstay of the economy, but losses in crop yield due to inadequate fertilization is still a major problem. Excessive fertilizer use can result in poor soil health and water pollution, whereas insufficient use can reduce crop yields and pose a risk to food security. The traditional approach to fertilizer recommendations is based on soil laboratory analysis and advice from an agronomist, which can be costly, time consuming and unavailable to smallholder farmers.

Machine learning methods have been extensively used for fertilizer recommendation based on soil, crop and environmental data, with the advent of IoT enabled sensors

and data driven farming. Most of the current models, however, are mainly concerned with prediction accuracy and are not interpretable, which makes them difficult to use in practice. Furthermore, the selection of fertilizers is a multi-criteria decision problem with trade-offs between yield, cost, environmental impact and soil suitability, which is not fully covered by the traditional classification methods.

To address these challenges, this paper introduces a hybrid XAI [1] approach combining Random Forest for fertilizer prediction, SHAP (SHapley Additive exPlanations) [2] for model interpretability, and Fuzzy TOPSIS [3] for multi-criteria decision-making. The proposed system offers an end-to-end pipeline that converts the raw soil and crop data into a ranked fertilizer recommendation and a human-readable explanation. This not only results in high predictive performance, but also in transparency and practical decision support for precision agriculture applications.

II. RELATED WORKS

Soil fertility prediction and fertilizer recommendation are two fields where machine learning and explainable artificial intelligence (XAI) have been extensively used in precision agriculture. Das *et al.* [4] emphasized the increasing significance of machine learning techniques in agricultural decision support systems, such as nutrient management and crop productivity enhancement. Abdullah *et al.* [5] evaluated several machine learning algorithms and concluded that ensemble models like Random Forest are better than traditional classifiers like SVM and SGD for soil fertility prediction tasks.

Although many machine learning models show high prediction accuracy, they are considered as black-box models, which makes them less interpretable in practical agricultural applications. Many machine learning models are considered as black-box models, and they show high prediction performance, but they are not interpretable in practical agriculture applications. To overcome this, Chandra

and Pawar [6] integrated the explainable AI techniques into soil fertility prediction models, enhancing transparency and user trust. Likewise, Wickramasinghe *et al.* [7] compared the interpretability of SHAP and LIME in the context of agriculture and found that SHAP offers more stable and reliable feature attribution explanations.

The problem of fertilizer recommendation is a multi-criteria decision making problem where there are trade-offs between crop yield, economic cost and environmental sustainability. Shivhare *et al.* [8] used the TOPSIS method to make fertilizer recommendations, and Faizi *et al.* [9] extended this concept to incorporate uncertainty into agricultural decision-making environments through the use of Fuzzy TOPSIS. Moreover, the emergence of intelligent smart farming systems is evident, as Acharya *et al.* [10] introduced an agentic AI-based approach to autonomous agricultural decision support.

However, previous research is largely limited to prediction, explanation or optimization individually. The proposed framework fills this gap by combining the Random Forest classification, SHAP explainability, and multi-criteria ranking using the Fuzzy TOPSIS method into an intelligent fertilizer recommendation system.

III. DATASET

A. Dataset Overview

The proposed framework is evaluated using a structured tabular dataset consisting of 10,000 agronomic records collected from the publicly available Kaggle platform [11]. Each record represents soil, crop and environmental condition that is mapped to one of seven fertilizer classes. The dataset contains 19 features and one target variable (Recommended_Fertilizer) which is a multi-class variable.

Among the input attributes, 12 are numerical attributes that represent soil properties (e.g., pH, moisture, nutrient levels), climatic conditions (e.g., temperature, humidity, and rainfall), and historical agricultural factors. The remaining 7 features are categorical, including soil type, crop type, crop growth stage, season, irrigation method, previous crop, and geographical region.

The Pearson correlation matrix of the numerical agronomic features is shown in Figure 1. The nutrient related variables (nitrogen, phosphorus, and potassium) have moderate correlations, and environmental attributes have comparatively weak correlations. The analysis provides useful insight into feature interactions and supports the feature engineering process used in the proposed framework.

B. Class Distribution

The target variable is highly imbalanced over the different fertilizer categories. Urea and DAP are the major classes while SSP is the least represented class. SMOTE [12] is used to

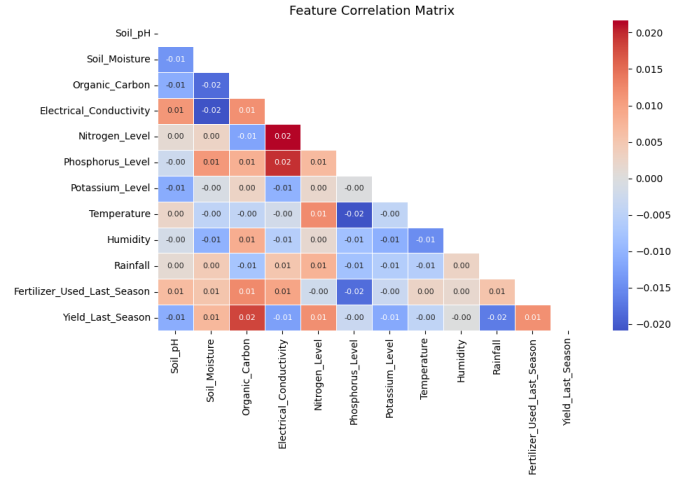


Fig. 1. Feature Correlation Matrix

balance this during training. The class-wise distribution is summarized in Figure 2.

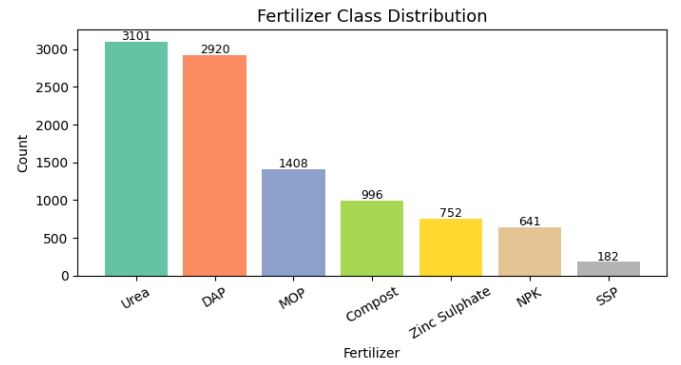


Fig. 2. Fertilizer Class Distribution

C. Feature Engineering

A number of features are engineered to enhance model performance and to capture the agronomic relationships that are not apparent in the data. The NPK ratio is the ratio of nitrogen to the total amount of phosphorus and potassium, which is a measure of the overall nutrient balance in the soil. Furthermore, a Nitrogen-Phosphorus interaction feature is created by multiplying the nitrogen and phosphorus levels, allowing the model to account for the interaction of the two main macronutrients. In addition, a soil-crop compatibility index is calculated from the interaction between soil type and crop type, which shows the compatibility of the soil and crop type.

These engineered features are added to the original features, and improve the model's ability to learn complex nonlinear relationships in the data.

IV. PROPOSED MODEL

The proposed framework consists of eight sequential stages, as shown in Figure 3. The system combines machine learning, explainable AI, and fuzzy multi-criteria decision-making to provide accurate and interpretable fertilizer recommendations. Each stage of the pipeline is described below.

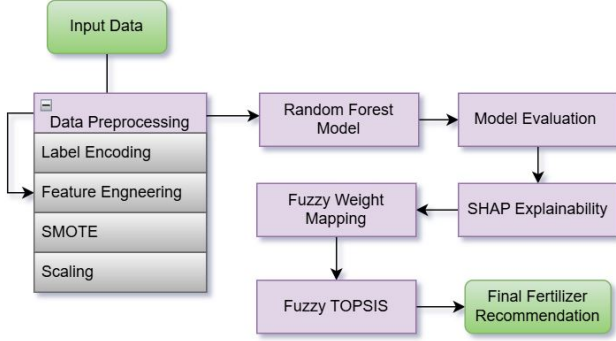


Fig. 3. Proposed Model

A. Data Preprocessing

The dataset contains both numerical and categorical attributes. The categorical features (Soil_Type, Crop_Type, Crop_Growth_Stage, Season, Irrigation_Type, Previous_Crop, Region) are encoded using Label Encoding. The target variable (Recommended_Fertilizer) is also converted to integer class labels.

A fixed random state is used to split the dataset into 80% train and 20% test sets, ensuring the results are reproducible. To overcome the class imbalance problem, SMOTE is only performed on the training set, which enhances the diversity of the training set without causing data leakage. All numerical features are scaled with StandardScaler fitted on the training data.

B. Random Forest Classifier

A Random Forest classifier with 300 decision trees is used as the primary predictive model. To address the imbalance that remains after SMOTE, the model is trained with `max_depth=None`, `min_samples_split=5`, `min_samples_leaf=2` and `class_weight='balanced'`.

Random Forest is selected because it is a good choice for structured agronomic data, robust to nonlinearity, and can be used with SHAP TreeExplainer for post-hoc interpretability. The model provides probability distributions of seven fertilizer classes. The architecture of the Random Forest model is shown in Figure 4.

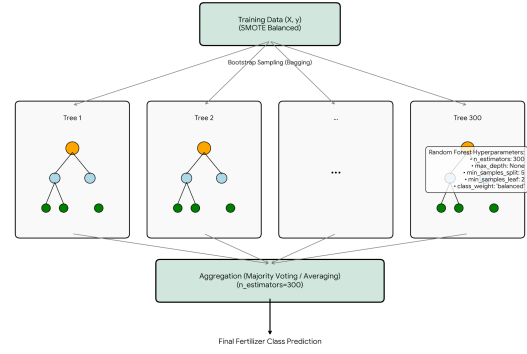


Fig. 4. Architecture of Random Forest

C. SHAP-Based Explainability

To improve interpretability, SHAP (SHapley Additive ex-Planations) is applied using TreeExplainer. SHAP assigns feature-wise contribution values to each prediction based on the Shapley value framework from cooperative game theory [2]. For a prediction $f(x)$, the SHAP value of feature i is defined as:

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f(S \cup \{i\}) - f(S)] \quad (1)$$

where F denotes the complete feature set and $S \subseteq F$ represents a subset excluding feature i . The formulation quantifies the marginal contribution of each feature to the final model prediction.

For global interpretability, the mean absolute SHAP value is computed across all samples and fertilizer classes:

$$\tilde{\phi}_i = \frac{1}{C} \sum_{c=1}^C \frac{1}{N} \sum_{j=1}^N |\phi_i^{(c,j)}| \quad (2)$$

where C is the number of fertilizer classes and N is the number of evaluated samples. This process determines the most influential agronomic characteristics such as nitrogen level, phosphorus level, potassium level and soil pH. Additionally, SHAP waterfall plots are provided to give local explanations for individual fertilizer recommendations.

D. Fuzzy Weight Mapping

The global SHAP importance scores are converted to fuzzy criteria weights to account for uncertainty in feature importance. The mean absolute SHAP values are first normalized to the range $[0,1]$ with min-max normalization:

$$\phi_i^{norm} = \frac{\phi_i - \phi_{min}}{\phi_{max} - \phi_{min} + \epsilon} \quad (3)$$

where ϕ_i represents the mean absolute SHAP value of feature i , and ϵ is a small constant used to avoid division by zero.

The normalized values are translated to Triangular Fuzzy Numbers (TFNs) with a linguistic scale of five levels: Very

Low, Low, Medium, High, and Very High. The levels are associated with a fixed fuzzy weight representation, which reflects the relative importance of the features.

These fuzzy weights are then passed as criteria importance values to the Fuzzy TOPSIS module, which allows for uncertainty-aware multi-criteria decision making.

E. Fuzzy TOPSIS Ranking

The Fuzzy TOPSIS module is used to rank fertilizer alternatives based on multiple agronomic criteria under uncertainty. The decision matrix is constructed using Triangular Fuzzy Numbers (TFNs) obtained from the SHAP-based fuzzy weight mapping. The distance between two triangular fuzzy numbers $\tilde{a}$ and $\tilde{b}$ is computed using the vertex method.

$$d(\tilde{a}, \tilde{b}) = \sqrt{\frac{(l_a - l_b)^2 + (m_a - m_b)^2 + (u_a - u_b)^2}{3}} \quad (4)$$

The separation from the fuzzy positive ideal solution (FPIS) and fuzzy negative ideal solution (FNIS) is given as:

$$d_i^+ = \sum d(\tilde{v}_{ij}, A^+), \quad d_i^- = \sum d(\tilde{v}_{ij}, A^-) \quad (5)$$

Finally, the closeness coefficient is computed as:

$$CC_i = \frac{d_i^-}{d_i^+ + d_i^-} \quad (6)$$

The alternatives are ranked in descending order of CC_i , and the highest-ranked fertilizer is selected as the final recommendation.

V. EXPERIMENTAL RESULTS AND ANALYSIS

A. Experimental Setup

The experiments are performed using Python 3.10 on Google Colab using an NVIDIA T4 GPU. Key library versions: scikit-learn 1.4, imbalanced-learn 0.12, shap 0.45, numpy 1.26, pandas 2.2, matplotlib 3.8, seaborn 0.13. Random seeds are set at 42 throughout for reproducibility. The dataset is divided into two parts: 80% for training (8,000) and 20% for testing (2,000), with stratification to ensure that the class proportions are maintained.

B. Classification Performance on Test Data

As shown in Table I, the Random Forest classifier is able to achieve good prediction accuracy on the test set that was not used for training on the multiclass fertilizer recommendation task. The overall accuracy of the model is 87.30% and macro-averaged F1-score is 74.96%, which shows balanced performance across all the fertilizer classes. The probabilistic output of the model allows uncertainty-aware decision support by giving class-wise confidence scores.

Table I presents the detailed classification report of the proposed fertilizer recommendation model, including precision, recall, F1-score, and support for each fertilizer class.

TABLE I
CLASSIFICATION REPORT FOR FERTILIZER RECOMMENDATION MODEL

Fertilizer Class	Precision	Recall	F1-score	Support
Compost	0.88	0.79	0.84	199
DAP	1.00	0.92	0.96	584
MOP	0.94	0.85	0.89	282
NPK	0.81	0.70	0.75	128
SSP	0.13	0.32	0.19	37
Urea	1.00	0.93	0.96	620
Zinc Sulphate	0.53	0.87	0.66	150
Macro avg	0.76	0.77	0.75	2000
Weighted avg	0.92	0.87	0.89	2000

To further analyze the classification performance across different fertilizer classes, the confusion matrix is presented in Figure 5.

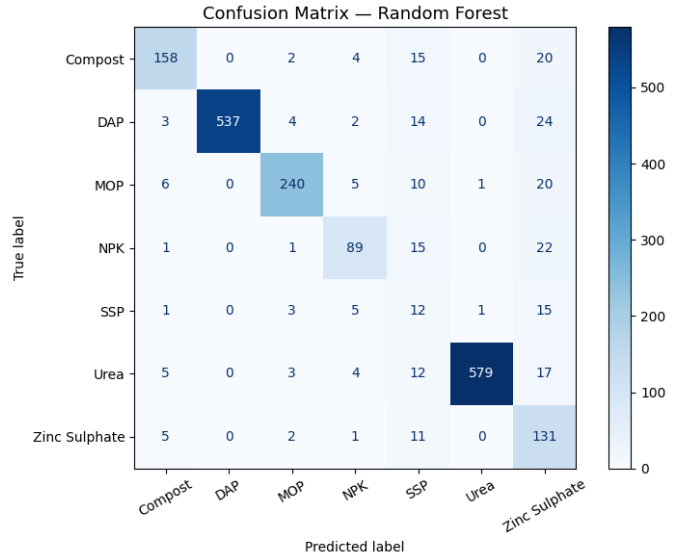


Fig. 5. Confusion matrix

C. Cross-Validation Performance

The Random Forest classifier is tested with Stratified 5-Fold Cross-Validation to guarantee robustness. The accuracy and macro-averaged F1-score are calculated over all folds.

The results show consistent performance across folds, indicating that the model is stable under different data partitions and not biased toward any specific subset. The bar chart in Figure 6 shows the accuracy and macro F1 score for each fold in the 5-Fold Cross-Validation.

D. SHAP-Based Explainability Analysis

SHAP TreeExplainer is used to quantify the contributions of features for both global and local explanations to improve interpretability.

The most influential predictors are identified in the global analysis (Figure 7 and Figure 8): Nitrogen Level, Phosphorus

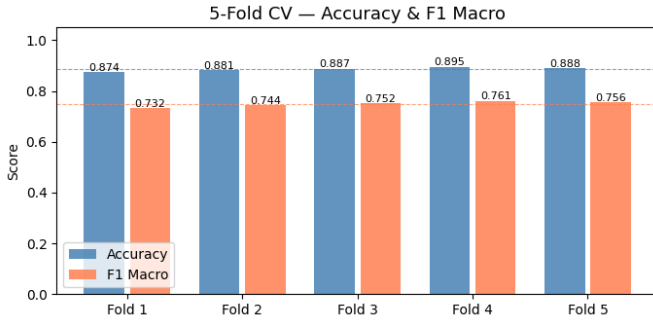


Fig. 6. 5-Fold Cross Validation

Level, Soil pH, and Potassium Level. The results are in line with agronomic principles, which state that the availability of macronutrients is the main factor affecting the need for fertilizers.

Other engineered features like NPK ratio, N-P interaction, and Soil crop index also feature in the top contributors, further substantiating the power of domain-driven feature engineering. Local SHAP waterfall analysis (Figure 9) provides additional insight into individual predictions by demonstrating the influence of each feature on the prediction towards or away from a particular fertilizer class.

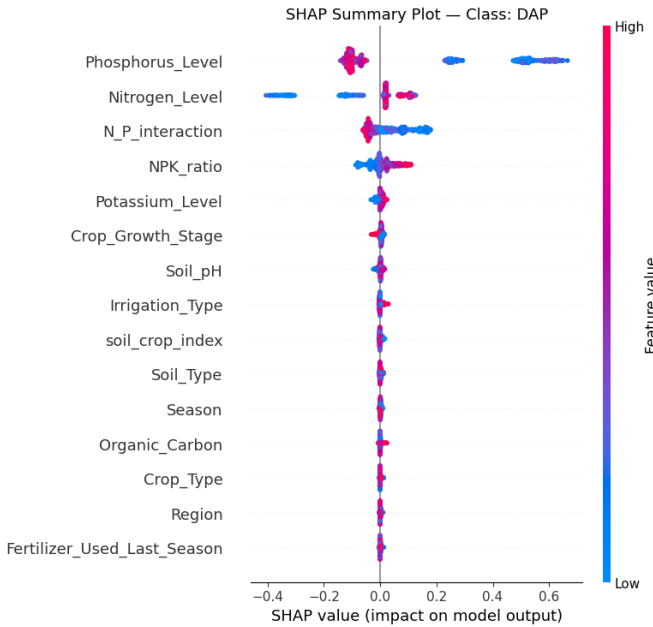


Fig. 7. SHAP Summary Plot — Class: DAP

E. Fuzzy Weight Transformation of SHAP Scores

The normalized SHAP scores are converted to Triangular Fuzzy Numbers (TFNs) with a linguistic mapping strategy (Very Low to Very High) to incorporate explainability into the

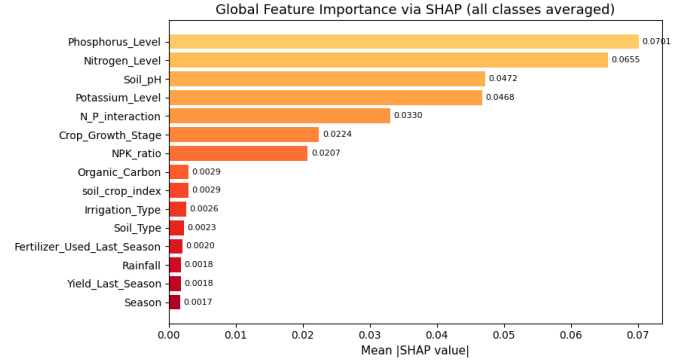


Fig. 8. Global Feature Importance via SHAP (all classes averaged)

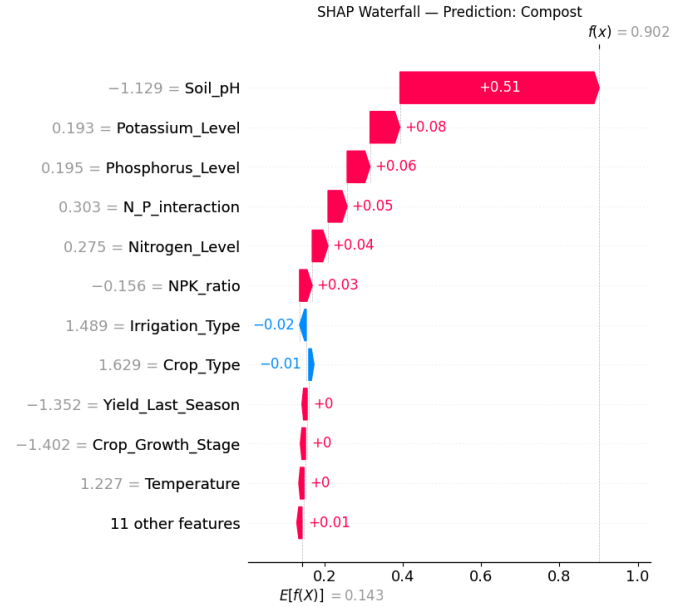


Fig. 9. SHAP Waterfall — Prediction: Compost

decision making process. These fuzzy weights are uncertainty-aware feature importance and applied in the TOPSIS stage. The mapping strategy is summarized in Table II.

TABLE II
LINGUISTIC MAPPING OF NORMALIZED SHAP SCORES TO TRIANGULAR FUZZY NUMBERS

Linguistic Term	TFN (l, m, u)	Normalized SHAP Range
Very High (VH)	(0.7, 0.9, 1.0)	≥ 0.75
High (H)	(0.5, 0.7, 0.9)	[0.50, 0.75]
Medium (M)	(0.3, 0.5, 0.7)	[0.25, 0.50]
Low (L)	(0.1, 0.3, 0.5)	[0.10, 0.25]
Very Low (VL)	(0.0, 0.1, 0.3)	< 0.10

This transformation enables bridging between machine learning explainability and fuzzy multi-criteria decision-making.

F. Fuzzy TOPSIS Ranking of Fertilizers

The fuzzy TOPSIS module is used to assess and rank the fertilizer alternatives based on the fuzzy weights obtained from the feature importance analysis based on SHAP. Multiple agronomic criteria are used to evaluate each fertilizer option, and the ranking is based on the Closeness Coefficient (CC) which is a measure of the relative distance of an alternative from the ideal best solution, but also the distance from the ideal worst solution. The ranking results computed are shown in Table III.

TABLE III
TOPSIS RANKING OF FERTILIZER ALTERNATIVES

Alternative	d^+	d^-	CC	Rank
Urea	3.526711	1.857291	0.344965	1
NPK	3.553640	1.868700	0.344630	2
DAP	3.531400	1.814655	0.339438	3
MOP	3.697974	1.587160	0.300307	4
Zinc Sulphate	3.756479	1.609493	0.299944	5
SSP	3.822079	1.495296	0.281209	6
Compost	3.817492	1.470293	0.278055	7

The experimental results show that Urea has the maximum Closeness Coefficient and is thus the best among all the fertilizer alternatives. NPK and DAP follow in the next rank, showing their high suitability in the soil and crop conditions. Compost, on the other hand, has the lowest CC value, indicating lower effectiveness with respect to the agronomic criteria evaluated. The overall results show that fuzzy TOPSIS analysis gives a systematic and interpretable multi-criteria decision making framework to obtain the most suitable fertilizer recommendation.

VI. FUTURE WORK

The proposed hybrid framework will be further developed for its robustness and applicability in real-world scenarios in future. Other agronomic parameters such as micronutrient content, pest status and current weather conditions can be incorporated to enhance the quality of the recommendations in dynamic field situations. The static agronomic rating matrix used in the Fuzzy TOPSIS module can also be replaced by expert-based or adaptive weightings to reflect the fertilizer suitability in the region. In addition, future studies can explore deep learning and federated learning techniques for large-scale distributed agricultural systems, maintaining data privacy. The integration of soil sensors with IoT and mobile decision-support platforms could also facilitate the development of fertilizer recommendation systems that are real-time and accessible to farmers.

VII. CONCLUSION

This paper proposed a hybrid explainable AI model for fertilizer recommendation, which combined Random Forest classification, SHAP interpretability, and Fuzzy TOPSIS multi-criteria decision-making. The system tackles the issues of prediction accuracy and decision transparency in precision agriculture.

The experimental results show that the Random Forest model has high classification accuracy and that the SHAP model can give meaningful explanations both globally and locally, identifying the most important agronomic factors, such as nitrogen, phosphorus, potassium and soil pH. In addition, the Fuzzy TOPSIS module allows multi-criteria ranking of the fertilizer alternatives using the fuzzy weights derived from the SHAP, which enhances the reliability of the decision beyond the single label classification.

The proposed framework provides accurate, interpretable and agronomically consistent fertilizer recommendations that can be used in real world agricultural decision support systems.

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